

CloudAge Global Services Pvt Ltd

Intelligent Financial Services AI Agent Platform

Enterprise Architecture Proposal for AWS Deployment

Prepared by: CloudAge Global Services Pvt Ltd

Date: June 2026

Document Version: 1.0

Classification: Confidential - Client Proposal

EXECUTIVE SUMMARY

CloudAge Global Services proposes the design, development, and deployment of an enterprise-grade intelligent financial services platform leveraging Amazon Bedrock AgentCore and Strands Agents SDK. This solution delivers production-ready AI agents capable of providing personalized financial guidance, budget management, and investment analysis at enterprise scale.

Business Objectives

- Deploy intelligent conversational agents for customer financial management
- Implement multi-agent orchestration for complex financial analysis
- Ensure enterprise-grade security, compliance, and observability
- Scale to support thousands of concurrent users with sub-second response times
- Reduce operational costs through intelligent automation

Key Deliverables

- Single-purpose Budget Assistant Agent with custom tools and visualizations
- Multi-Agent System with specialized Budget and Financial Analysis Agents
- Orchestrator Agent for intelligent query routing and response synthesis
- Production deployment on AWS with complete observability stack
- Enterprise security framework with guardrails and compliance controls

SOLUTION OVERVIEW

Platform Capabilities

Our proposed solution delivers a comprehensive AI-powered financial advisory platform with the following capabilities:

1. Personalized Financial Guidance

- Tailored recommendations based on income and spending patterns
- Context-aware responses across multiple interactions
- Responsible AI with built-in guardrails and disclaimers

2. Intelligent Budget Management

- Automated 50/30/20 budget rule calculations
- Spending pattern analysis and categorization
- Visual analytics with pie charts and spending visualizations
- Real-time budget tracking and alerts

3. Investment Analysis & Portfolio Management

- Market research and analysis
- Portfolio performance tracking
- Risk assessment and recommendations
- Investment opportunity identification

4. Enterprise-Grade Features

- Conversation memory and state management
- Structured outputs for consistent, parseable financial reports
- Multi-channel deployment support (web, mobile, messaging platforms)
- Integration with existing financial systems and APIs

TECHNICAL ARCHITECTURE

Architecture Overview

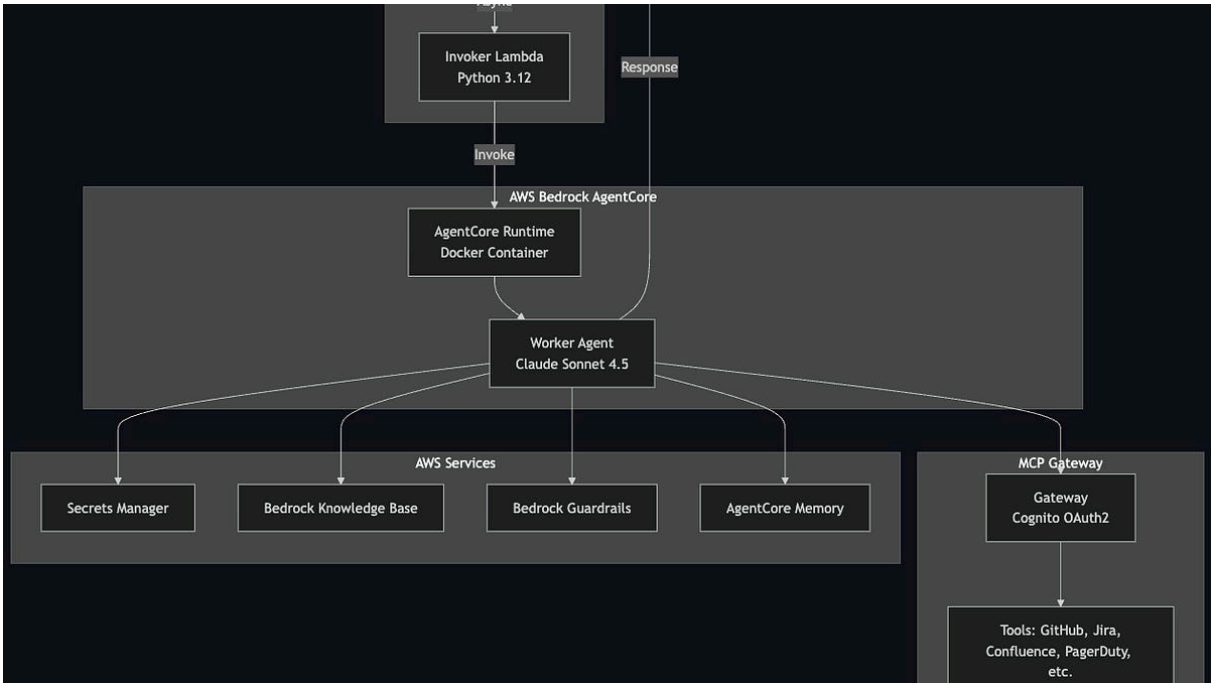


Figure 1: High-level architecture showing AgentCore integration with AWS services

Single Agent Architecture

The Budget Agent implementation includes:

Core Components:

- **AgentCore Runtime:** Docker containerized execution environment
- **Foundation Model:** Claude Sonnet 4.5 via Amazon Bedrock
- **Custom Tools:** Financial calculations, data visualization, report generation
- **Memory Layer:** Conversation context and state persistence
- **Security Layer:** IAM policies, guardrails, and access controls

AWS Services Integration:

- **Secrets Manager:** Secure credential management
- **Bedrock Knowledge Base:** Financial regulations and best practices
- **Bedrock Guardrails:** Content safety and compliance enforcement
- **AgentCore Memory:** Session and conversation state management
- **CloudWatch:** Comprehensive monitoring and logging

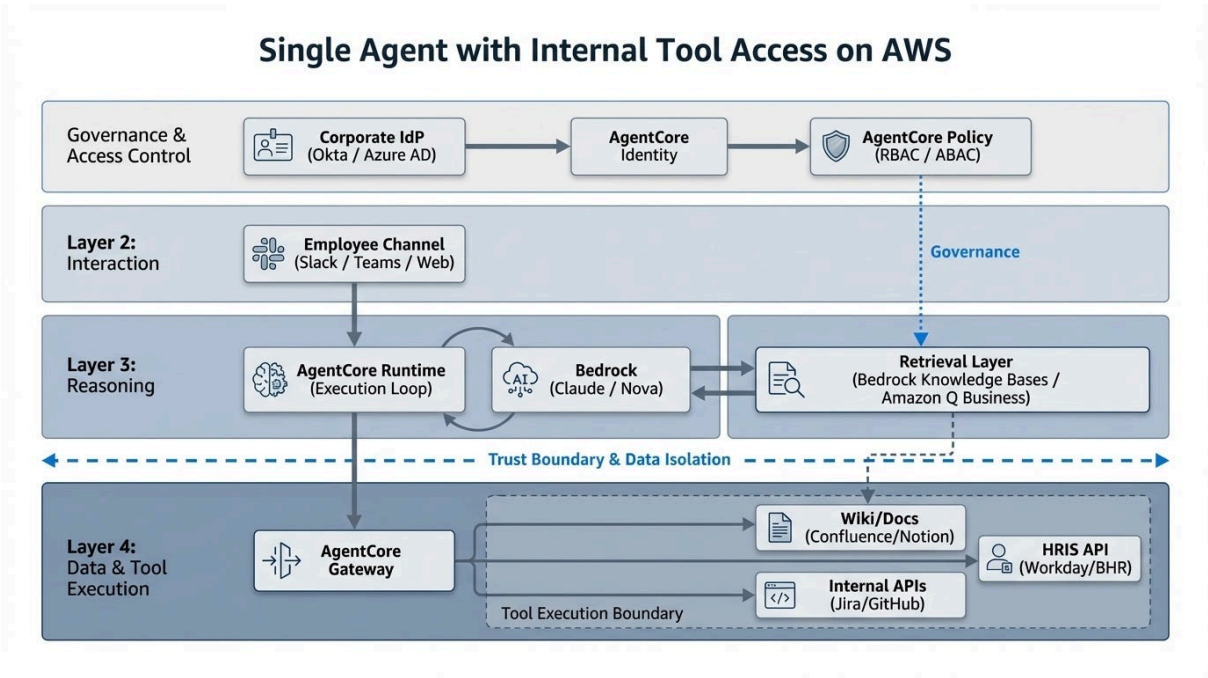


Figure 2: Budget Agent architecture with custom tools and memory

Multi-Agent Architecture

Our multi-agent system employs the "Agents as Tools" pattern with specialized components:

Specialized Agents:

1. Budget Agent

- Personal budgeting expertise
- Spending analysis and categorization
- Financial discipline recommendations
- Budget optimization strategies

2. Financial Analysis Agent

- Investment research and analysis
- Portfolio management guidance
- Market trend analysis
- Risk assessment and mitigation

3. Orchestrator Agent

- Intelligent query routing
- Multi-agent coordination
- Response synthesis and validation
- Quality assurance and compliance checking

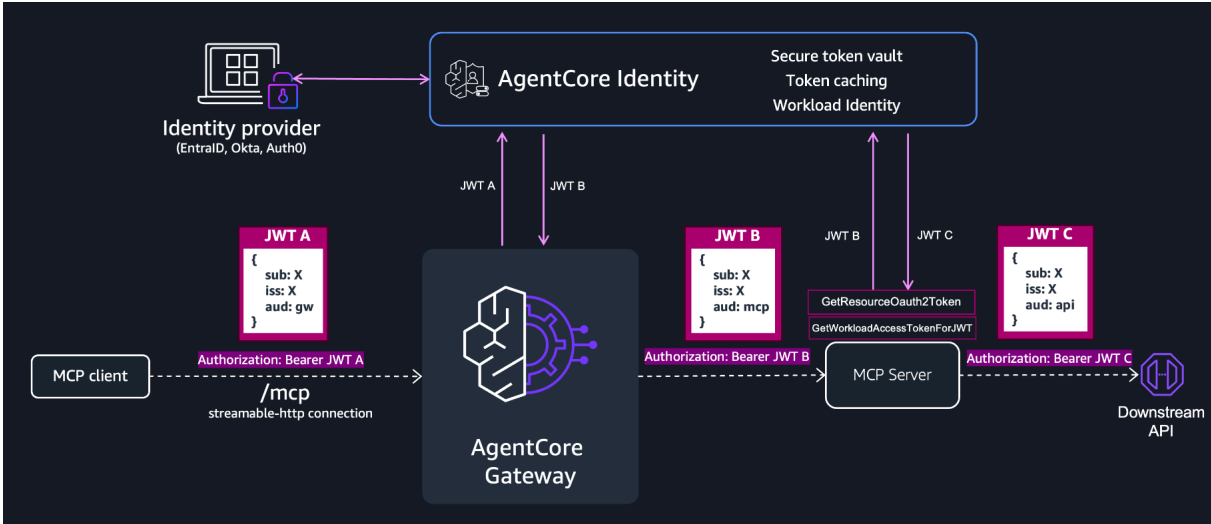


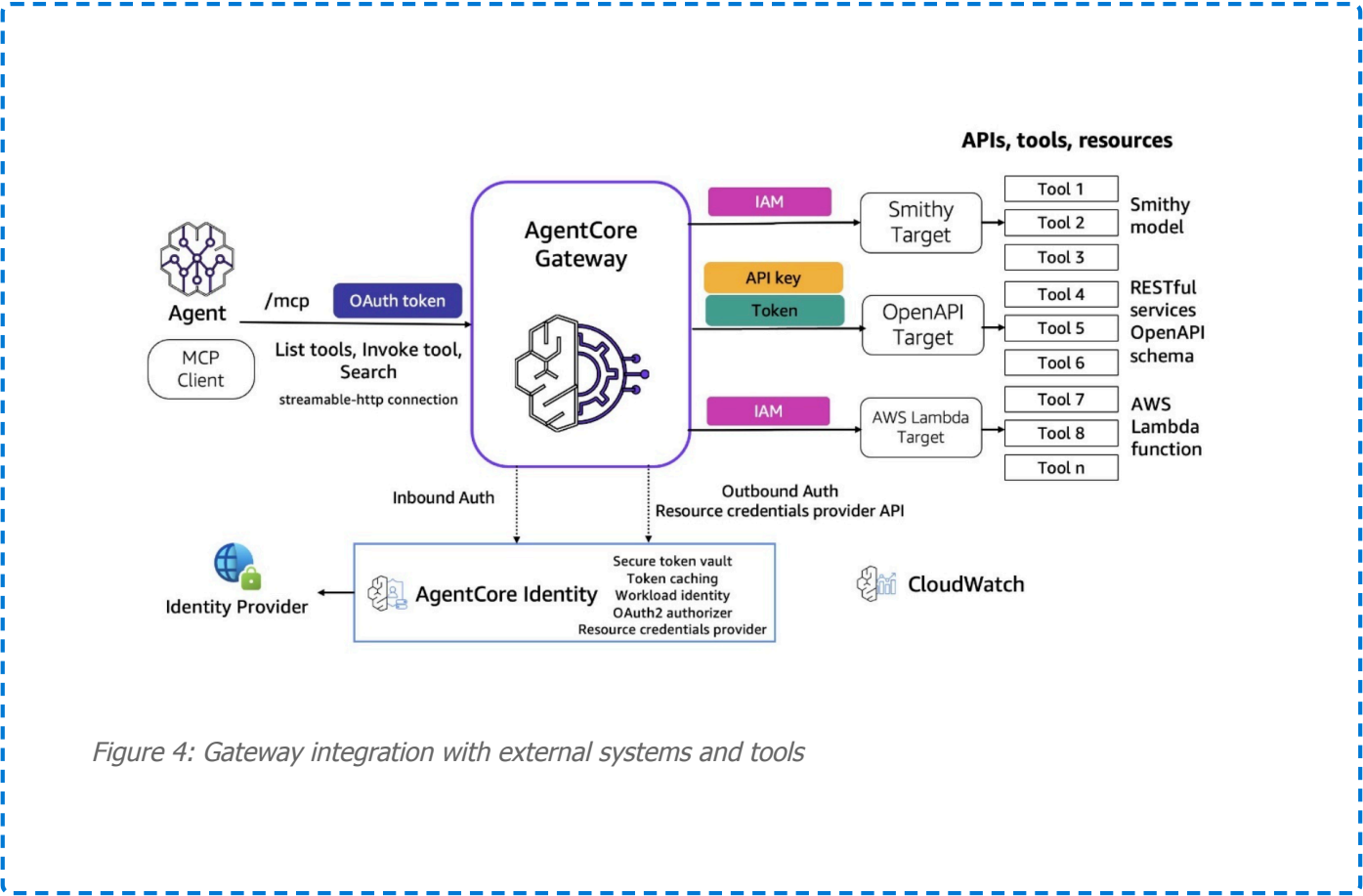
Figure 3: Multi-agent system with orchestrator and specialized agents

AgentCore Gateway Integration

The AgentCore Gateway provides secure tool integration and API connectivity:

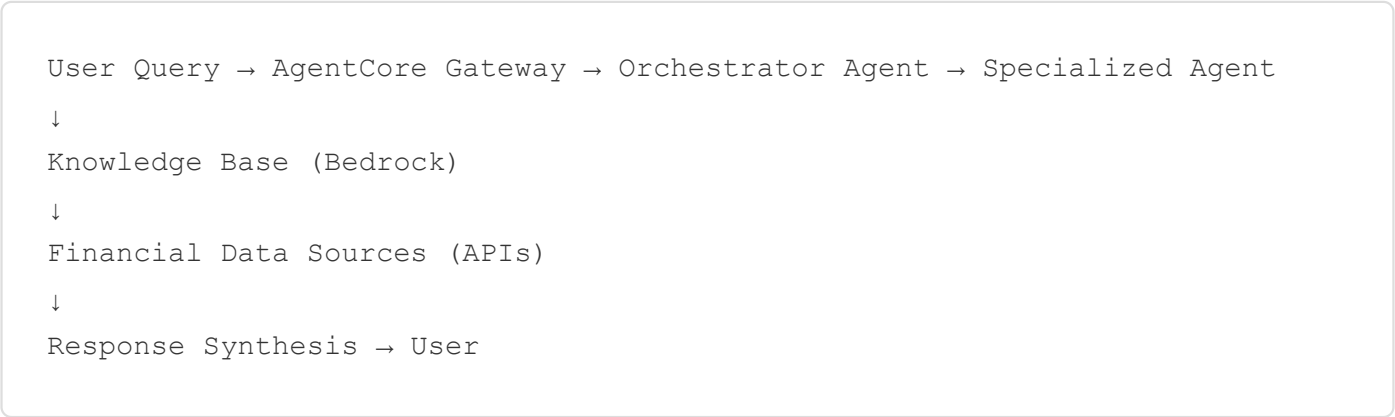
Integration Capabilities:

- **MCP (Model Context Protocol):** Standardized agent-to-tool communication
- **OAuth2 Authentication:** Secure identity management
- **API Gateway:** RESTful API endpoints for external integrations
- **Lambda Functions:** Serverless compute for custom business logic
- **Smithy Targets:** Type-safe API integrations



Data Architecture

Data Flow:



Data Storage:

- **Amazon S3:** Document storage and knowledge base
- **DynamoDB:** Session state and conversation history
- **ElastiCache (Redis):** Semantic caching and fast lookups
- **Aurora Serverless:** Structured financial data and analytics

IMPLEMENTATION APPROACH

Phase 1: Foundation & Single Agent Deployment (Weeks 1-4)

Objectives:

- Establish AWS infrastructure and security baseline
- Deploy AgentCore Runtime environment
- Build Budget Agent with core functionality
- Implement custom tools for financial calculations

Deliverables:

- Infrastructure as Code (CloudFormation/Terraform)
- Budget Agent with 50/30/20 rule implementation
- Visualization tools (pie charts, spending analytics)
- Conversation memory implementation
- Basic monitoring and logging

Milestones:

- Week 1: Infrastructure setup and security configuration
- Week 2: AgentCore deployment and foundation model integration
- Week 3: Custom tool development and testing
- Week 4: Budget Agent deployment and validation

Phase 2: Multi-Agent System Development (Weeks 5-8)

Objectives:

- Develop Financial Analysis Agent
- Implement Orchestrator Agent with intelligent routing
- Build inter-agent communication protocols
- Establish response synthesis and validation

Deliverables:

- Financial Analysis Agent with investment capabilities

- Orchestrator Agent with routing logic
- Agent communication framework
- Enhanced monitoring for multi-agent interactions

Milestones:

- Week 5: Financial Analysis Agent development
- Week 6: Orchestrator Agent implementation
- Week 7: Integration testing and optimization
- Week 8: Multi-agent system validation

Phase 3: Enterprise Integration & Security (Weeks 9-12)

Objectives:

- Implement enterprise security controls
- Integrate with existing financial systems
- Deploy guardrails and compliance frameworks
- Establish observability and alerting

Deliverables:

- Security framework with IAM policies and encryption
- API integrations with financial data providers
- Bedrock Guardrails for content safety
- CloudWatch dashboards and alerting
- Audit logging and compliance reporting

Milestones:

- Week 9: Security implementation and penetration testing
- Week 10: External system integrations
- Week 11: Guardrails and compliance validation
- Week 12: Observability stack deployment

Phase 4: Production Deployment & Optimization (Weeks 13-16)

Objectives:

- Production environment deployment
- Performance optimization and load testing

- User acceptance testing
- Knowledge transfer and documentation

Deliverables:

- Production-ready deployment
- Performance benchmarks and optimization report
- User acceptance test results
- Operations runbook and documentation
- Training materials and knowledge transfer sessions

Milestones:

- Week 13: Production environment setup
- Week 14: Load testing and performance tuning
- Week 15: User acceptance testing
- Week 16: Go-live and handover

SECURITY & COMPLIANCE

Security Architecture

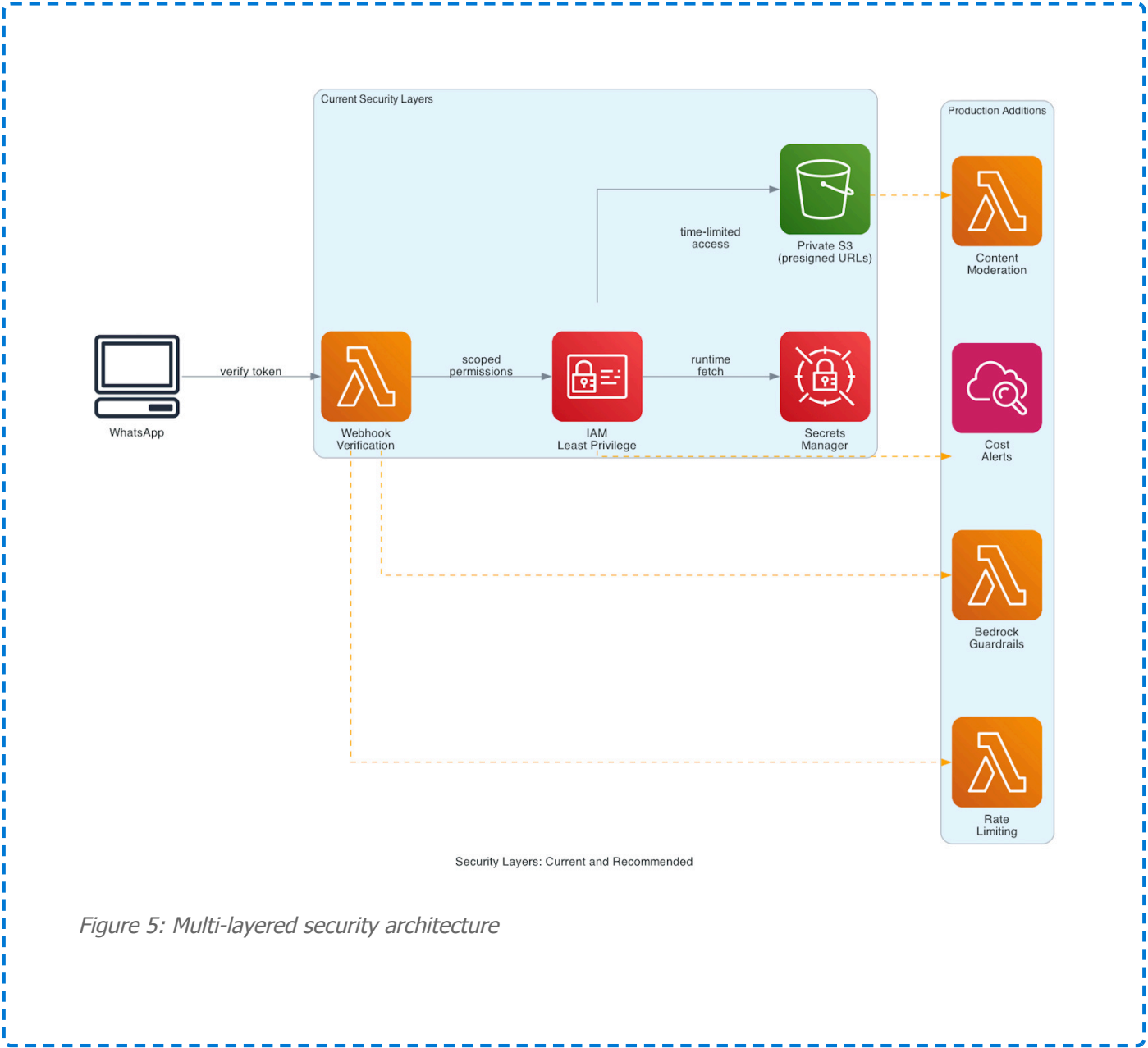


Figure 5: Multi-layered security architecture

Security Controls:

1. Identity & Access Management

- AgentCore Identity with OAuth2 integration
- Role-based access control (RBAC)
- Attribute-based access control (ABAC)
- Multi-factor authentication (MFA)

2. Data Protection

- Encryption at rest (AES-256)
- Encryption in transit (TLS 1.3)
- Token-based authentication
- Secure secrets management

3. Network Security

- VPC isolation with private subnets
- Security groups and NACLs
- AWS WAF for web application protection
- DDoS protection with AWS Shield

4. Application Security

- Bedrock Guardrails for content moderation
- Input validation and sanitization
- Rate limiting and throttling
- Automated security scanning

Compliance Framework

Regulatory Compliance:

- SOC 2 Type II
- ISO 27001
- GDPR data protection
- PCI DSS for payment data (if applicable)
- Financial services regulations (as applicable)

Audit & Monitoring:

- Comprehensive audit logging
- Real-time security event monitoring
- Automated compliance reporting
- Incident response procedures

DEPLOYMENT STRATEGY

Infrastructure Architecture

AWS Services Utilized:

Category	Services
Compute	AWS Lambda, ECS/Fargate, EC2 (if needed)
AI/ML	Amazon Bedrock, Bedrock AgentCore
Storage	S3, DynamoDB, Aurora Serverless, ElastiCache
Security	IAM, Secrets Manager, KMS, GuardDuty
Networking	VPC, API Gateway, CloudFront, Route 53
Monitoring	CloudWatch, CloudTrail, X-Ray
Integration	EventBridge, SQS, SNS

High Availability Design

Multi-Region Deployment:

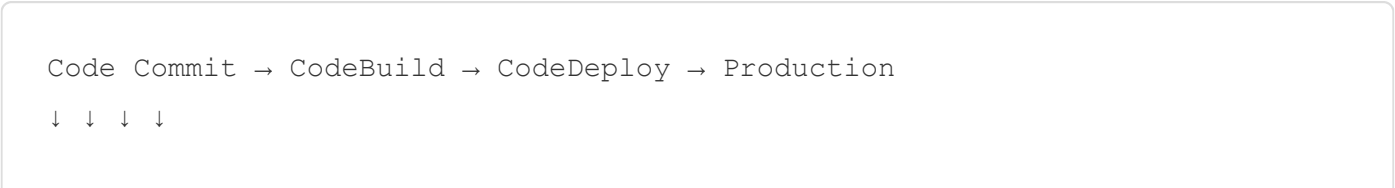
- Primary region: Active production workload
- Secondary region: Standby for disaster recovery
- RPO (Recovery Point Objective): < 15 minutes
- RTO (Recovery Time Objective): < 1 hour

Scalability:

- Auto-scaling based on demand
- Serverless architecture for elastic scaling
- Caching layer for performance optimization
- Database read replicas for query scaling

CI/CD Pipeline

Deployment Automation:



Testing	Build & Staging	Monitoring
Package Validation		

Tools:

- AWS CodePipeline for orchestration
- AWS CodeBuild for compilation and testing
- AWS CodeDeploy for automated deployments
- Infrastructure as Code (CloudFormation/Terraform)

OBSERVABILITY & MONITORING

Monitoring Stack

Key Metrics Tracked:

- Agent response time and latency
- Tool execution success rate
- Error rates and exception tracking
- Resource utilization (CPU, memory, storage)
- Cost metrics and budget tracking
- User engagement and satisfaction

CloudWatch Dashboards:

- Real-time operational dashboard
- Business metrics dashboard
- Cost optimization dashboard
- Security and compliance dashboard

Logging Strategy

Log Levels:

- **Application Logs:** Agent decisions, tool calls, errors
- **Access Logs:** API calls, authentication events
- **Audit Logs:** Security events, compliance tracking
- **Performance Logs:** Latency metrics, resource usage

Log Retention:

- Operational logs: 30 days
- Audit logs: 7 years (compliance requirement)
- Performance logs: 90 days

Alerting Framework

Alert Categories:

- **Critical:** Service outage, security breach, data loss
- **High:** Performance degradation, error rate spikes
- **Medium:** Resource utilization thresholds
- **Low:** Informational events, routine maintenance

Notification Channels:

- Email for low/medium priority
- SMS/PagerDuty for critical/high priority
- Slack/Teams integration for operations team

PERFORMANCE SPECIFICATIONS

Service Level Objectives (SLOs)

Metric	Target	Measurement
Availability	99.9% uptime	Monthly rolling average
Response Time	< 2 seconds	95th percentile
Error Rate	< 0.1%	Percentage of failed requests
Throughput	1000 concurrent users	Peak capacity
Data Freshness	< 5 minutes	For real-time data

Capacity Planning

Initial Deployment:

- Concurrent users: 1,000
- Daily active users: 10,000
- Monthly active users: 50,000
- API calls per day: 500,000

Scalability Targets:

- 10x growth capacity built-in
- Auto-scaling triggers at 70% utilization
- Database scaling to 100TB+
- Cache scaling based on hit ratio

COST ANALYSIS

Estimated Monthly Costs (Production)

AWS Infrastructure Costs:

Service	Estimated Cost (USD)
Amazon Bedrock	\$2,500 - \$5,000
AgentCore Runtime	\$500 - \$1,000
Lambda Functions	\$200 - \$500
DynamoDB	\$300 - \$600
S3 Storage	\$100 - \$300
Aurora Serverless	\$400 - \$800
ElastiCache	\$200 - \$400
CloudWatch & Monitoring	\$150 - \$300
Data Transfer	\$200 - \$500
Security Services	\$300 - \$600
Total Estimated	\$4,850 - \$10,000

Cost Optimization Strategies:

- Reserved capacity for predictable workloads (30-40% savings)
- Spot instances for non-critical batch processing
- Intelligent caching to reduce API calls
- Automated scaling to match demand patterns
- Regular cost optimization reviews

Total Cost of Ownership (3-Year Projection)

Year	Infrastructure	Development	Operations	Total
Year 1	\$120,000	\$200,000	\$80,000	\$400,000
Year 2	\$150,000	\$50,000	\$100,000	\$300,000

Year	Infrastructure	Development	Operations	Total
Year 3	\$180,000	\$50,000	\$120,000	\$350,000
Total	\$450,000	\$300,000	\$300,000	\$1,050,000

RISK MANAGEMENT

Risk Assessment & Mitigation

Risk	Probability	Impact	Mitigation Strategy
Model accuracy issues	Medium	High	Continuous evaluation, human-in-the-loop validation, A/B testing
Security breach	Low	Critical	Multi-layer security, regular audits, penetration testing, incident response plan
Compliance violations	Low	High	Automated guardrails, compliance monitoring, legal review
Performance degradation	Medium	Medium	Auto-scaling, caching, performance testing, CDN
Cost overruns	Medium	Medium	Budget alerts, cost monitoring, optimization reviews
Vendor lock-in	Low	Medium	Multi-model support, abstraction layers, portable architecture

Business Continuity

Disaster Recovery:

- Multi-region deployment
- Automated backup and recovery
- Regular DR testing (quarterly)
- Documented runbooks for failover

Data Protection:

- Automated backups (hourly incremental, daily full)
- Point-in-time recovery capability
- Cross-region replication
- Data retention policies

GOVERNANCE & OPERATIONS

Operating Model

Support Structure:

- **Level 1:** Automated self-service, chatbot support
- **Level 2:** Technical support team (business hours)
- **Level 3:** CloudAge engineering team (24/7 for critical issues)
- **Escalation:** AWS Premium Support partnership

Maintenance Windows:

- Scheduled maintenance: Sunday 2:00 AM - 4:00 AM UTC
- Emergency patches: As needed with 4-hour notice
- Feature deployments: Blue-green deployment (zero downtime)

Change Management

Change Control Process:

1. Change request submission
2. Impact assessment and approval
3. Testing in non-production environment
4. Scheduled deployment with rollback plan
5. Post-deployment validation
6. Documentation update

Version Control:

- Git-based source control
- Semantic versioning
- Release notes and changelogs
- Rollback capability to previous versions

SUCCESS METRICS & KPIs

Technical KPIs

Metric	Target	Measurement Frequency
System availability	99.9%	Real-time
Average response time	< 2 seconds	Hourly
Error rate	< 0.1%	Daily
Mean time to recovery (MTTR)	< 15 minutes	Per incident
Deployment frequency	Weekly	Weekly
Change failure rate	< 5%	Monthly

Business KPIs

Metric	Target	Measurement Frequency
User adoption rate	70% of target users	Monthly
Customer satisfaction (CSAT)	> 4.5/5.0	Quarterly
Query resolution rate	> 85% first contact	Weekly
Cost per query	< \$0.01	Monthly
Automation rate	> 80% automated	Monthly

Financial KPIs

Metric	Target	Measurement Frequency
ROI	200% within 18 months	Quarterly
Cost savings vs. manual	60% reduction	Quarterly
Infrastructure cost efficiency	< \$0.005 per query	Monthly

PROJECT TEAM & RESPONSIBILITIES

CloudAge Team

Role	Responsibilities	Allocation
Project Manager	Overall project delivery, stakeholder management	Full-time
Solution Architect	Architecture design, technical oversight	50%
Lead ML Engineer	Agent development, model optimization	Full-time
DevOps Engineer	Infrastructure, CI/CD, deployment	Full-time
Security Specialist	Security implementation, compliance	50%
QA Engineer	Testing, quality assurance	Full-time

Client Team Requirements

Role	Responsibilities
Product Owner	Requirements, priorities, acceptance criteria
Technical Lead	Integration points, API access, data sources
Security Liaison	Security reviews, compliance validation
Business SME	Financial domain expertise, validation

DELIVERABLES

Technical Deliverables

1. Infrastructure

- Complete AWS infrastructure deployment
- Infrastructure as Code templates
- Network architecture documentation
- Security configuration documentation

2. Application Components

- Budget Agent with custom tools
- Financial Analysis Agent
- Orchestrator Agent
- AgentCore Gateway configuration
- API documentation

3. Data & Integration

- Database schemas and migrations
- API integration specifications
- Data mapping documents
- ETL pipelines (if applicable)

4. Security & Compliance

- Security architecture document
- Compliance assessment report
- Penetration test results
- Audit logging configuration

Documentation Deliverables

1. Technical Documentation

- Solution architecture document
- API reference documentation
- Deployment runbooks
- Operations manual
- Disaster recovery procedures

2. User Documentation

- End-user guide
- Administrator guide
- Troubleshooting guide
- FAQ and knowledge base

3. Training Materials

- Technical training for operations team
- End-user training materials
- Video tutorials
- Hands-on lab exercises

ACCEPTANCE CRITERIA

Phase Acceptance Criteria

Phase 1 (Single Agent):

- ☐ Budget Agent successfully deployed and operational
- ☐ 50/30/20 budget calculations accurate
- ☐ Visualization tools functional
- ☐ Conversation memory working across sessions
- ☐ Basic monitoring dashboards operational
- ☐ Security controls implemented and tested

Phase 2 (Multi-Agent):

- ☐ Financial Analysis Agent deployed and functional
- ☐ Orchestrator Agent routing accuracy > 95%
- ☐ Inter-agent communication stable
- ☐ Response synthesis quality validated
- ☐ Performance benchmarks met

Phase 3 (Security & Integration):

- ☐ All security controls validated
- ☐ Penetration testing passed
- ☐ Compliance requirements met
- ☐ External integrations functional
- ☐ Guardrails preventing policy violations

Phase 4 (Production):

- ☐ Production deployment successful
- ☐ Load testing passed (1000 concurrent users)
- ☐ User acceptance testing approved
- ☐ Documentation complete
- ☐ Knowledge transfer completed

Final Acceptance Criteria

- ☐ All SLOs met or exceeded
- ☐ Security audit passed
- ☐ Performance benchmarks achieved

- ☐ User acceptance sign-off
- ☐ Operations team trained and certified
- ☐ All documentation delivered and approved

SUPPORT & MAINTENANCE

Post-Deployment Support

Support Periods:

- **Warranty Period:** 90 days post-go-live (included)
- **Standard Support:** Year 1-3 (optional)
- **Premium Support:** 24/7 coverage (optional)

Support Services:

- Bug fixes and patches
- Performance optimization
- Security updates
- Minor feature enhancements
- Technical support and troubleshooting

Service Level Agreements (SLAs)

Severity	Response Time	Resolution Time	Availability
Critical (P1)	15 minutes	4 hours	24/7/365
High (P2)	1 hour	8 hours	Business hours + on-call
Medium (P3)	4 hours	24 hours	Business hours
Low (P4)	1 business day	5 business days	Business hours

Maintenance Services

Proactive Maintenance:

- Weekly health checks
- Monthly performance reviews
- Quarterly security assessments
- Annual architecture review
- Cost optimization reviews (monthly)

Updates & Upgrades:

- Security patches: Within 48 hours of release
- Feature updates: As per agreed roadmap

- Model updates: Quarterly or as needed
- Infrastructure upgrades: Scheduled maintenance windows

ASSUMPTIONS & DEPENDENCIES

Assumptions

1. Client Responsibilities:

- Timely access to required systems and APIs
- Availability of subject matter experts for requirements clarification
- Prompt review and approval of deliverables
- Provision of test data and scenarios

2. Technical Assumptions:

- AWS account with appropriate permissions available
- Network connectivity and firewall rules can be configured
- Third-party API integrations have available documentation
- Existing systems can support integration requirements

3. Resource Assumptions:

- Client technical team available for knowledge transfer
- Decision-makers available for timely approvals
- Budget approved for AWS infrastructure costs
- Compliance and security teams available for reviews

Dependencies

1. External Dependencies:

- AWS service availability in target region
- Third-party API availability and rate limits
- Financial data provider integrations
- Regulatory approval (if required)

2. Internal Dependencies:

- Client infrastructure team for network setup
- Client security team for compliance reviews
- Client business team for UAT participation
- Client operations team for handover

TERMS & CONDITIONS

Project Terms

- **Project Duration:** 16 weeks from contract signing
- **Payment Terms:** 30% upfront, 40% at Phase 2 completion, 30% at final acceptance
- **Warranty:** 90 days post-deployment for bug fixes
- **Intellectual Property:** Client owns custom code; CloudAge retains framework IP
- **Confidentiality:** Mutual NDA in effect

Change Management

- Changes to scope require written change order
- Impact assessment provided within 5 business days
- Changes may affect timeline and cost
- Emergency changes handled through expedited process

Termination

- Either party may terminate with 30 days written notice
- Work completed and expenses incurred billed up to termination date
- Knowledge transfer and documentation provided upon termination
- IP rights as per original agreement maintained

NEXT STEPS

Immediate Actions Required

1. Client Review (Week 1):

- Review and approve this proposal
- Identify project team members
- Schedule kickoff meeting

2. Contract Finalization (Week 2):

- Negotiate and sign Master Services Agreement
- Define Statement of Work (SOW)
- Establish payment terms and schedule

3. Project Initiation (Week 3):

- Conduct project kickoff meeting
- Set up communication channels and tools
- Grant AWS access and credentials
- Begin Phase 1 discovery and planning

Contact Information

CloudAge Global Services Pvt Ltd

Project Inquiries:

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Technical Questions:

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Phone: +91-XXX-XXX-XXXX

Account Management:

Email: accounts@cloudage.com

Phone: +91-XXX-XXX-XXXX

APPENDICES

Appendix A: Technology Stack

Development Stack:

- Python 3.12+
- Strands Agents SDK
- LangChain (optional)
- Docker for containerization
- Git for version control

AWS Services:

- Amazon Bedrock (Claude Sonnet 4.5, Nova)
- Amazon Bedrock AgentCore
- AWS Lambda
- Amazon DynamoDB
- Amazon Aurora Serverless
- Amazon ElastiCache (Redis)
- Amazon S3
- Amazon API Gateway
- Amazon CloudWatch
- AWS Secrets Manager
- AWS Identity and Access Management (IAM)
- AWS Key Management Service (KMS)

Monitoring & Observability:

- Amazon CloudWatch
- AWS X-Ray
- CloudTrail
- Custom dashboards and alerts

Appendix B: Compliance Certifications

CloudAge maintains the following certifications:

- ISO 27001:2022
- SOC 2 Type II
- AWS Advanced Consulting Partner

- AWS Machine Learning Competency
- GDPR compliant operations

Appendix C: References & Case Studies

Available Upon Request:

- Financial services client case studies
- Technical reference architectures
- Security whitepapers
- Performance benchmark reports

ABOUT CLOUDAGE GLOBAL SERVICES

CloudAge Global Services Pvt Ltd is a leading cloud consulting firm specializing in AI/ML solutions, cloud architecture, and digital transformation. With deep expertise in AWS technologies and a proven track record in financial services, we deliver enterprise-grade solutions that drive business value.

Our Capabilities:

- AI/ML Solution Development
- Cloud Architecture & Migration
- DevOps & Site Reliability Engineering
- Security & Compliance
- Data & Analytics
- Digital Transformation

Why Choose CloudAge:

- AWS Advanced Consulting Partner
- 100+ successful AI/ML deployments
- Deep financial services domain expertise
- 24/7 global delivery capabilities
- Commitment to innovation and excellence

Document Prepared By:

CloudAge Global Services Pvt Ltd

Date: June 2026

Version: 1.0

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End of Proposal Document